

AI in Healthcare SoC'26

Week 1 Assignment

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1 5 interesting concepts from lectures

1. Levels of Computer Vision

One of the interesting concepts I learned was that computer vision can be divided into low-level, mid-level, and high-level tasks. Low-level vision focuses on basic image processing operations such as resizing, edge detection, and segmentation. Mid-level vision involves understanding relationships between images, such as reconstructing 3D scenes, estimating motion, and creating panoramas. High-level vision deals with semantic understanding, including image classification, object detection, and scene interpretation.

2. Human Vision Inspired Computer Vision

I learned that many computer vision techniques are inspired by the way humans see and interpret the world. The human eye captures light and converts it into electrical signals, which are then processed by the brain to recognize objects, colors, and motion. Similarly, cameras capture visual information through image sensors, and computer algorithms process this data to identify patterns and extract useful information.

3. Image Interpolation

I found image interpolation particularly interesting because it explains how images can be resized without appearing excessively pixelated. When an image is enlarged, new pixel values must be estimated from existing ones. Techniques such as bilinear interpolation calculate a weighted average of neighboring pixels to create smooth transitions, while triangular interpolation uses geometric relationships between nearby pixels to estimate values. These methods significantly improve image quality during scaling operations and are widely used in image editing software, computer graphics, and computer vision applications.

4. Color Spaces

Another concept I learned was the importance of color spaces in digital imaging. A color space provides a mathematical framework for representing colors so that they can be stored, processed, and displayed consistently across different devices. The sRGB color space is one of the most widely used standards for digital images and displays. Learning about color spaces helped me understand how computers represent colors numerically and why color management is necessary to ensure that images appear similar on cameras, monitors, and other devices.

5. Images as Matrices of Pixel Values

I learned that a digital image can be represented as a matrix in which each element corresponds to a pixel. For grayscale images, each pixel stores a single intensity value, while color images store separate values for red, green, and blue channels. This mathematical representation allows computers to perform operations such as filtering, enhancement, compression, and object detection directly on image data.

2 Healthcare Reports

Healthcare reports are documents used to record, communicate, and analyze patient information, diagnoses, treatments, and medical procedures. They are essential for maintaining accurate medical records and ensuring effective communication among healthcare professionals.

2.1 Types of Healthcare Reports

Some common types of healthcare reports include:

- **Patient Medical History Reports** - Contain information about a patient's previous illnesses, surgeries, medications, and family medical history.
- **Laboratory Reports** - Present the results of medical tests such as blood tests, urine tests, and pathology examinations.
- **Discharge Summaries** - Summarize a patient's hospital stay, diagnosis, treatment, and follow-up instructions.
- **Operative Reports** - Document details of surgical procedures performed on patients.
- **Medical Imaging Reports** - Describe findings from imaging examinations such as X-rays, CT scans, MRI scans, ultrasound scans, and PET scans.

3 Medical Imaging Reports

Medical imaging reports are specialized healthcare documents prepared by radiologists after analyzing medical images. Their primary purpose is to communicate imaging findings to physicians and other healthcare professionals for diagnosis and treatment planning.

3.1 Components of a Medical Imaging Report

A typical medical imaging report contains:

- **Patient Information:** Name, age, gender, and identification details.
- **Examination Details:** Type of imaging procedure performed and the body part examined.
- **Clinical History:** Relevant symptoms or medical concerns that led to the imaging study.

- **Findings:** Detailed observations made from the medical images.
- **Impression/Conclusion:** A concise summary of the most important findings and possible diagnosis.
- **Recommendations:** Suggestions for follow-up tests, additional imaging, or treatment if required.

3.2 Importance of Medical Imaging Reports

Medical imaging reports play a vital role in modern healthcare because they:

- Help physicians make accurate diagnoses.
- Assist in monitoring disease progression and treatment effectiveness.
- Provide a permanent record of imaging findings.
- Improve communication between radiologists and other healthcare professionals.
- Support informed clinical decision-making and patient care.

3.3 Challenges

Creating medical imaging reports requires expertise and precision. Ambiguous language, incomplete findings, or inconsistent report structures can lead to misunderstandings and potentially affect patient care. To address these challenges, modern healthcare systems increasingly use artificial intelligence and natural language processing techniques to assist radiologists in generating accurate and standardized reports.